

Incource-1

CSE-1201

1st year 2nd Semester(2023-24)

Date: 30/12/2025

Time: 45 mins

Marks: 20

1. What are C Tokens? Write down the structure of a C program. 4
2. Mention the operators used in C. 2
3. Describe basic data types with examples. 4
4. Write down the syntax with flow chart for WHILE DO and FOR statements. 6
5. Write a C program using the else if ladder statement. 4

Department of Computer Science and Engineering

Incource-2

CSE-1201

1st year 2nd Semester Examination-2024 (Session: 2023-24)

Date: 26/01/2026

Time: 45 mins

Marks: 20

1. Define Array with it's classifications. 4
2. Write a C program to find the factorial value of 7 using recursive function. 4
3. In the following fragment of code what is the final value of x and y? 2
 - i) x=10; y=++x;
 - ii) x=5; y=x++;
 - iii) x=4; y=--x;
4. Differentiate between Structure and Array. 3
5. Write a C program to input and print array elements using Pointer. 4
6. What will be the output of the following program segment? 3

```
int is_even(int num) {
    if (num % 2 == 0) {
        return 1;
    } else {
        return 0;
    }
}

int main() {
    int num = 5;
    if (is_even(num)) {
        printf("%d is even\n", num);
    } else {
        printf("%d is odd\n", num);
    }
    return 0;
}
```

Mymensingh Engineering College
Department of Computer Science and Engineering
CSE 1202: Digital Logic Design

Class Test 1

Time: 40 minutes

Marks: 10

1. Design an OR gate using NAND gate and AND gate Using NOR gate.
2. Minimize the following boolean function using K-map-
 $F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$
3. Design the logic circuit for the expression-
 $AB'C + A'BC + ABC'$

4

4

2

Mymensingh Engineering College
Department of Computer Science and Engineering
CSE 1202: Digital Logic Design

Class Test 2

Time: 45 minutes

Marks: 15

5. Draw and describe the circuit diagram of J-K flipflop with truth table.
6. Draw the block diagram and truth table for BCD to 7 segment display.
7. Draw the block diagram of a 16X1 MUX
8. What are the differences between latch and flipflop?

5

5

3

2

Mymensingh Engineering College
Department of Computer Science and Engineering
1st year 2nd Semester
PHY - In course - 01

Time: 1 Hour

Total Marks - 25

1. (a) Define simple harmonic motion and explain its characteristics. 3
(b) Show that the average kinetic energy and average potential energy is same and constant and half of total energy. 7
(c) A body is vibrating with SHM of amplitude 15 cm and frequency 4112. Compute the maximum Acceleration and velocity. 3
2. (a) Explain the terms Resonance and Stationary wave. 4
(b) Obtain an expression for the differential equation of a plane progressive wave. 4
(c) Explain the terms Node and Antinode. 4

Mymensingh Engineering College
Department of Computer Science and Engineering
1st year 2nd Semester 2024
PHY - 1203, In course: 02

Time: 1 Hour

Total Marks - 25

- (a) State and explain zeroth law of Thermodynamics and first Law of Thermodynamics. 4
- (b) Define c_p and c_v . For adiabatic process prove that $TV^{\gamma-1} = \text{constant}$. 6
- (c) What is Efficiency of Heat Engine? Find out its Efficiency. 4
- (d) Define (i) coherent source (ii) Huygens principle. 4
- (e) What is interference? Find the condition for Constructive and Destructive Interference. 7

Answer three (3) questions from the following (You should answer all subsections of the same question consecutively.)		10x2=20
1.	a. Evaluate (any two): i) $\int \sqrt{x} (\ln x)^2 dx$ ii) $\int \tan^{-1} \frac{1-x^2}{1+x^2} dx$ iii) $\int e^x \frac{1+2x+4x^2}{(1+x)^2} dx$	4
	b. Use the Maclaurin series for $\cos x$ to obtain the Maclaurin series for $x \cos 2x$	3
	c. Examine the convergency (any three): i) $\int_2^{\infty} \frac{dx}{\sqrt{x^2-1}}$ ii) $\int_1^2 \frac{x^2 dx}{(2-x)^2}$ iii) $\int_0^{\infty} \frac{\sqrt{x} dx}{(1+x)}$ iv) $\int_0^{\infty} \frac{\sin x dx}{x}$ v) $\int_0^{\infty} (1-e^{-x}) \frac{\cos x}{x^2} dx$	3
2.	a. Evaluate $\int e^{-x^2} dx$ as an infinite series.	2
	b. Find the first four terms of the Taylor's series for the following functions: i) $\ln x$ centered at $a=1$ ii) $\frac{1}{x}$ centered at $a=1$ iii) $\sin x$ centered at $a=\frac{\pi}{6}$	4
	c. Evaluate (any two): i) $\int \frac{(x^2+x+1) dx}{(x^2+x^2-6x)}$ ii) $\int_2^3 \frac{(1-x^2) dx}{(x^2-1)(2x+1)}$ iii) $\int x e^x \cos x dx$	4
3.	a. Find the differential equation of all circles passing through origin and having their centers on the x-axis.	2
	b. From an ODE from $xy^2 = ax + b \ln x$ where a, b are constants and identify it.	2
	c. Calculate the value of the integral $\int_1^{3.2} \log x dx$ i) Simpson's $\frac{1}{3}$ rule ii) Simpson's $\frac{3}{8}$ rule and also find the errors.	6
4.	a. Define singular point and improper integral of second kind.	2
	b. Form a differential equation for a cardioid $r = a(1 + \cos \theta)$	2
	c. Solve (any three): i) $(y^2 + x^2) dx = 2xy dy$ ii) $\frac{dy}{dx} = \frac{1}{x} + \tan \frac{y}{x}$ iii) $\frac{dy}{dx} = \frac{2(2x-y)^{-2}}{3x-y+4}$ iv) $(1+e^x) dx + e^{\frac{x}{2}} (1-e^{\frac{x}{2}}) dy = 0$ iv) $(x \sin \frac{y}{x} - y \cos \frac{y}{x}) dx + x \cos \frac{y}{x} dy = 0$ and also find the order and degree of the corresponding problems.	6
5.	a. State and prove the necessary and sufficient condition of exactness.	5
	b. Evaluate (any two): i) $\int_{-\infty}^{\infty} \frac{x dx}{x^4+1}$ ii) $\int_1^2 \frac{dx}{(x+1)\sqrt{(x^2-1)}}$ iii) $\int \frac{(1-\cos \theta) d\theta}{\cos \theta (1+\cos \theta)}$	4
	c. Write short notes (any two): i) ODE ii) PDE iii) Linear-Nonlinear differential equations	1

Answer two (2) questions from the following (You should answer all subsections of the same question consecutively.)		10x2=20
1.	a. Find the value of x for which the series of $x - \frac{x^2}{2^2} + \frac{x^3}{3^2} - \frac{x^4}{4^2} + \dots \dots \dots \infty$	5
	b. Find the whole perimeter of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$	5
	c. Discuss the convergence of the series $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2+1}$	3
2.	a. Find the whole arc length of the cycloid $x = a(\theta + \sin \theta), y = a(1 - \cos \theta)$	5
	b. Evaluate (any one): i) $\int \frac{(3 \sin x - 4 \cos x - 5) dx}{(2 + \cos x - 2 \sin x)}$ ii) $\int \frac{dx}{a \cos x + b \sin x}$	2
3.	a. Test the convergence and divergence of the series of $5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \dots \dots \dots$	4
	b. Write short notes (any two): i) Alternating Series ii) Power Series iii) sequence iv) series	4
	c. Examine the nature of the series $1 + 2 + 3 + 4 + \dots \dots \dots + \infty$	2
4.	a. Test the series for convergence and absolute convergence of $1 - \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} - \frac{1}{4\sqrt{4}} + \dots \dots \dots \infty$	5
	b. Find the surface area of the solid generated by the revolution of the cardioid $r = a(1 + \cos \theta)$ about the initial line.	5

Myensing Engineering College
Department of Computer Science and Engineering
2nd In-course Exam' 2024
Sub: Developing English Skill, Course code: 1205
Marks - 20 Time - 45 minutes

1. Use appropriate prepositions in the blanks: 1x5=5

- Physical exercise is beneficial — health.
- He was sick — over eating.
- Put — your shoes.
- Rahim looks — his parents.
- The girl takes — her mother.

2. Transform the sentence as directed: 1x5=5

- Without working hard, you cannot succeed in life. (make it compound)
- A rolling stone gathers no moss (complex).
- Only he can do it. (negative)
- Iron is one of the most useful metals. (Positive)
- He is as strong as a lion. (Comparative)

3. Write a paragraph about "The problems of Technical Education in Bangladesh!" 10

Department of Computer Science & Engineering
1st year 2nd semester In-course Exam' 2025
Subject: Developing English Language Skills
Subject code: 1205
Total marks - 20 Time - 30 minutes

1. Fill in the blanks with suitable article. Use cross(x) mark where an article is not necessary. 5

(a) — idle man and (b) — active man cannot be equal.
We know (c) — story of an ant and (d) — grasshopper.
An ant was industrious. On the other hand, the grasshopper was really (e) — lazy.

2. Fill in the gaps with correct form of verbs as given in the box below: 5

Bangladesh is full of natural resources. The prosperity of the country (a) — on the proper utilization of the resources. We should not (b) — a lazy life. We should all (c) — up our country. For this reason, we have to (d) — hard. No nation can prosper without industry. It should be (e) — that industry is the key to success. If we (f) — the days away, we (g) — behind. The nations that (h) — industries (i) — the pinnacle of development. So we (j) — them.

build	idle	be	lag	lead
remember	work	reach	depend	follow

3. Write a paragraph of 100 words about "Importance of Artificial Intelligence in Education sector?" 10